

MC-AB-SFC2: A MASQ-Based Replication of a Multi-Country Agent-Based Stock-Flow Consistent Model

Malcom Chumchoua Penda

University of Douala

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1. Purpose and Patterns

1.1. Model Purpose

The purpose of the Multi-Country Agent-Based Stock-Flow Consistent (MC-AB-SFC) model is to analyze the impact of different fiscal regimes on the long-term economic dynamics of a Monetary Union broadly comparable to the European Economic and Monetary Union (EMU) (Caiani et al., 2018).

1.2. Replication Purpose

The purpose of this work is to replicate the core mechanisms of the MC-AB-SFC model developed by Caiani et al. (2018), while adopting a modular and test-driven engineering approach based on the MASQ metamodel (Dinu et al., 2012; Stratulat et al., 2009). This replication aims to provide a reproducible, and **extensible implementation** of the model while maintaining consistency with its original theoretical framework.

To improve readability and ensure consistency with the implementation adopted in this replication, some notational conventions used in the original article have been modified. These changes are purely notational and do not alter the economic mechanisms, accounting system and behavioral assumptions of the original model (see section 7.1 for more details).

1.3. Patterns

As the original model, this replicated model aims to reproduce the following economic patterns within monetary union:

- Real GDP and labor productivity growth

- Unemployment and inflation dynamics
- Public debt and deficit dynamics;
- International trade dynamics;
- R&D investments dynamics;
- real and financial cycle dynamics:
- heterogeneous macroeconomic countries dynamics.
- firm and bank size distributions;

2. Entities, State Variables, and Scales

2.1. Entities and states variables

The original model description identifies explicitly two types of entities: agents and markets. Following the MASQ approach, the replicated model distinguishes five types of entities represented in Figure 1:

- *agents* which represent economic agents as decision-makers,
- *spaces* which represent environments where economic interactions take place,
- *roles* which represent agents' behavior during interactions within spaces,
- *stocks* which represent items created and exchange during interactions.

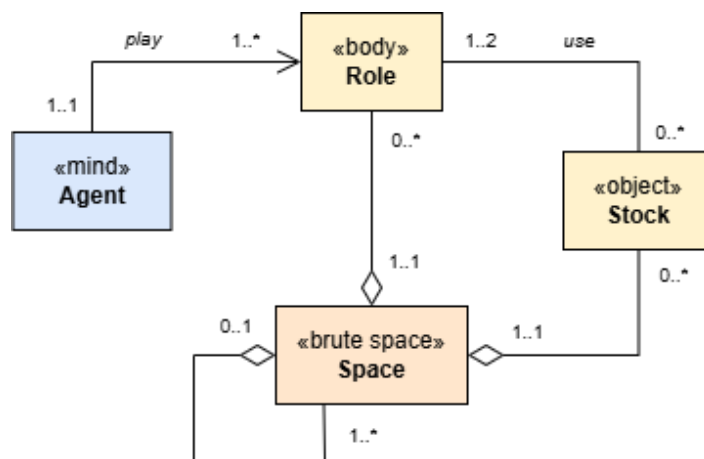


Figure 1. Types of entities

This novel entity taxonomy constitutes an interpretation of the original model and is not intended to introduce additional economic structures or mechanism. This taxonomy was partly

inspired by roles and stocks identified by Caiani and Caverzasi (2017) and Seppecher (2011) in their AB-SFC models.

2.1.1. Agent types and states variables

The modelled economy comprises five types of agents identified in the original model :

- *households* who search jobs, consume goods and invest equity,
- *firms* who organize production and R & D investment,
- *banks* which supply credit to firms and manage deposits account,
- *governments* who define fiscal and budget policy of countries
- *central banks* who define and implement monetary policy.

Each agent has an *internal state* which the state of its memory (Dinu et al., 2012). This internal state is made of expectations, decisions and lagged indicators. The main decisions variables of agent internal state are presented in Table 1 while other variables of agent internal state will be described in appropriate sub-models of agent behaviors.

Table 1. Agents' state decisions variables

Agent	Symbols	Variables	Domains	Descriptions
Household	E_{ht}^D	equity_star	$\mathbb{R}$	equities desired
	D_{ht}^D	deposit_star	$\mathbb{R}^+$	deposits desired
	w_{ht}^D	reservation_wage	$\mathbb{R}^+$	reservation wage
	C_{ht}^D	cons_star	$\mathbb{R}^+$	desired consumption level
Firm	w_{it}	offered_wage	$\mathbb{R}^+$	wage offered
	p_{it}	offered_price	$\mathbb{R}^+$	price offered
Bank	L_{jt}^{DS}	credit_max	$\mathbb{R}^+$	maximum amount of credit
	r_{jt}^D	deposit_rate	$\mathbb{R}^+$	deposit interest rate
Central Bank	r_{nt}	discount_rate	$\mathbb{R}^+$	discount interest rate
Government	G_{kt}	expenditures	$\mathbb{R}^+$	level of public expenditures
	τ_{kt}	tax_rate	$\mathbb{R}^+$	tax rate

Finally, firms and central banks are divided into several categories:

- *tradable Firms* who produce goods which are tradable internationally;
- *non-tradable Firms* who produce goods which are not tradable internationally;
- *national central banks* who interact with national commercial banks or government;
- *union central bank* who defines monetary policy of the union.

2.1.2. Space types and states variables

The model contains two geographical spaces:

- *countries* which represent domestic economies where national transactions take place;
- *monetary union* which represents the common environment which enable international interactions.

These geographical spaces encompass six types of market (Caiani et al., 2018):

- national *labor markets*,
- national *deposit markets*,
- national *good markets* for non-tradable goods,
- a common *good market* for tradable goods,
- a common *credit market*,
- a common *bond market*.

Some spaces maintain shared state variables that characterize the conditions of agent's decisions and interactions. Theses shared state variables are presented in Table 2.

Table 2. Spaces with shared state variables

Spaces	Symbol	Variable	Domain	Description
Goods market	P_t	avg_price	$\mathbb{R}^+$	average price
	Φ_t	avg_prod	$\mathbb{R}^+$	average level of productivity
Labor market	u_t	unemployment	$\mathbb{R}^+$	unemployment rate
Country	ΔP_t	inflation	$\mathbb{R}$	level of inflation
	Y_{kt}	gdp	$\mathbb{R}^+$	aggregate GDP
	$P\Gamma_t^{default}$	prob_failure	$\mathbb{R}^+$	probability of company failure

2.1.3. Role types and states variables

Roles allow concretely agents to interact in specific markets or geographical spaces. A role links an agent to a space and provide method for agent perception and action. The Table 3 presents roles types with their relationships with spaces and agents:

Table 3. Roles types and relationships description

Spaces	Roles	Agents	Economic Functions
<i>Goods market</i>	<i>Producer</i>	Firms	goods supply
	<i>Consumer</i>	Households	goods demand
<i>Labor market</i>	<i>Worker</i>	Households	labor supply
	<i>Employer</i>	Firms	labor demand
<i>Credit market</i>	<i>Lender</i>	Banks	credit supply
	<i>Borrower</i>	Firms	credit demand
<i>Deposit market</i>	<i>Depositor</i>	Households/Firms	deposit supply
	<i>Deposit Bank</i>	Banks	deposit demand
	<i>Deposit Guarantee</i>	Governments	deposit insurance
<i>Bond market</i>	<i>Bond Issuer</i>	Governments	bonds supply
	<i>Bond Buyer</i>	Banks/Central Banks	bonds demand
<i>Country</i>	<i>Citizen</i>	Households	household transfers
	<i>Company</i>	Firms/Banks	profit distribution
	<i>Fiscal Authority</i>	Government	budget policy
<i>Monetary Union</i>	<i>Monetary Authority</i>	Central Banks	monetary policy
	<i>Reserve Holder</i>	Banks/Governments	reserve payments

The role state is made of indicators of their agents previous or current activity. This approach is inspired by the definition of Tranier (2007) of *dynamic state* of objects in environment. Indeed, the dynamic state of an object is the set of variables indicating activities of this object.

So, monetary or fiscal authority roles expose results of decision process of central banks and government. Similarly, within markets, some roles expose prices of their demand or supply. Other roles expose indicators of financial wealth. These exposed variables form the dynamic state of these roles as shown in Table 4:

Table 4. Roles with dynamic state variables

Stocks	Symbol	Variable	Domain	Description
Producer	p_t	price	$\mathbb{R}^+$	price offered
	ϕ_t	productivity	$\mathbb{R}^+$	productivity
Employer	w_t	wage	$\mathbb{R}^+$	wage offered
Borrower	L_t^D	credit_demand	$\mathbb{R}^+$	credit demand
	A_t	net_worth	$\mathbb{R}^+$	net worth indicator
Bond Issuer	B_t/Y_t	debt_ratio	$\mathbb{R}^+$	debt-to-GDP ratio
Fiscal Authority	τ_t	tax_rate	$\mathbb{R}^+$	tax rate exposed
Monetary Authority	r_t	discount_rate	$\mathbb{R}^+$	discount rate exposed

2.1.4. Stock types and state variables

Stocks represent any object created, exchanged and destroyed by economic transactions. Stocks are contained within specific space where they are created. The stocks identified in the model are described in Table 5:

Table 5. Stocks types and spaces description

Spaces	Stocks	Economic meaning
Goods market	<i>Inventory</i>	unsold goods accumulated by a firm
	<i>Sale</i>	good sold by a firm to a household
Labor market	<i>Job</i>	labor supplied by a household to a firm
Credit market	<i>Loan</i>	credit supplied by a bank to a firm
	<i>Advance</i>	cash advance of a central bank to a bank
Deposit market	<i>Deposit</i>	bank deposit held by a firm or a household
Bond market	<i>Bond</i>	government debt held by an economic agent
Country	<i>Equity</i>	equities held by a household and issued by a firm or bank
	<i>Transfer</i>	cumulated transfers between a government and an agent
Monetary Union	<i>Reserve</i>	reserves account held by a government or a central bank

These stocks allow to register flows which are changes in stocks over time. The state of each stock is made of the accumulation of flows. So, state variables of modeled stocks are described in Table 6

Table 6. Stocks dynamic state variables

Stocks	Symbol	Variable	Domain	Description
Bond	B_t	principal	$\mathbb{R}^+$	bonds principal value
	Int_t^B	interests	$\mathbb{R}^+$	interests paid on bond
Deposit	D_t	amount	$\mathbb{R}^+$	deposit amount
	Int_t^D	interests	$\mathbb{R}^+$	interests paid on deposit
Loan	L_t	principal	$\mathbb{R}^+$	loan amount
	Int_t^L	interests	$\mathbb{R}^+$	interests paid on loan
Advance	A_t	principal	$\mathbb{R}^+$	advance value
	Int_t^A	interests	$\mathbb{R}^+$	interests paid on advance
Sale	$\hat{q}_t$	quantity	$\mathbb{R}^+$	quantity sold
	p_t	price	$\mathbb{R}^+$	selling price
Inventory	inv_t	quantity	$\mathbb{R}^+$	level of inventories
	q_t	quantity_prod	$\mathbb{R}^+$	quantity produced
Job	l_t	labor_sold	$\mathbb{R}$	quantity of labor sold
	w_t	wage_paid	$\mathbb{R}^+$	amount of wage paid
Equity	E_t	amount	$\mathbb{R}^+$	equity value
	Div_t	dividends	$\mathbb{R}^+$	dividends paid
Transfer	T_t	taxes_paid	$\mathbb{R}^+$	taxes paid
	G_t	public_transfers	$\mathbb{R}^+$	public transfer paid
	π_t	profit_transfers	$\mathbb{R}^+$	profits transferred
Reserve	R_t	amount	$\mathbb{R}^+$	reserves amount
	Int_t^R	interests	$\mathbb{R}^+$	interests paid on reserves

The state of each stock is linked to the stocks and flows of SFC accounting matrices (see section 7.2 for details). Indeed, each stock correspond to an associated balance-sheet position and each flow of the transaction matrix can be computed with the state of modeled stock. This intends to preserves stock-flow consistency.

2.2. Scales description

The model has not measurable agents' space necessary for economic interactions. Indeed, agents has no positions in model spaces. Even in the original model, the Salop circle represent agent preferences not agent spatial positions.

The time period modeled represents a quarter as suggested in original article (Caiani et al., 2018). Similarly, the model runs for 1000 periods.

3. Process overview and scheduling

A round of simulation represent a time period. During each round of the simulation, in compliance with the original article, the following sequence of events takes place:

1. **Production and R&D investment planning:** Firms determine their desired production, their labor demand, the price of their output, the wage offered, and their desired R&D investment.
2. **Credit markets matching:** Firms interact with banks on the credit market and possibly receive loans. Banks possibly ask cash advances to the Central Bank to satisfy the mandatory liquidity ratio.
3. **Labor markets matching:** Firms interact with workers on the labor market.
4. **Production, R&D and incomes distribution:** Workers are paid and employed to produce firms' output and to perform R&D. Dividends generated in the previous period are distributed to equity holders, summing up to their current income.
5. **Tax collection and public expenditures:** Governments calculate revenues from taxes (on past period profits and current period households' income), determine the level of public spending and the tax rate for the next period, repay bonds plus interests to bond holders, and determine the quantity of bonds to be issued.

6. **Bond markets matching:** Bonds are put on the bond market where commercial banks buy it. The possible residual part is purchased by national Central Banks.
7. **Goods consumption:** After having paid taxes and received the tax-exempt monetary transfer from the government, households compute their demand for consumption goods and interact with tradable and non-tradable firms on the correspondent good markets.
8. **Profit distribution planning:** Firms and banks compute their profits and update their net worth and shareholders' equity accordingly. Taxes and dividends to be paid in the next period to the government and to equity holders, respectively, are then computed.
9. **Enter-exit:** Defaulted firms and banks exit the market. Household equity investment takes place and, if enough financial resources are collected, new firms and banks are created.

4. Design Concepts

4.1. Basic Principles

4.2. Emergence

4.3. Adaptation

4.4. Objectives

4.5. Learning

Learning is not implemented

4.6. Prediction

4.7. Sensing

4.8. Interaction

4.9. Stochasticity

4.10. Collectives

The model includes no collectives.

4.11. Observation

5. Initialization

6. Input Data

No input data is used.

7. Submodels

7.1. Notation conventions and deviations from the original model

The mathematical notation used in this document follows the notation introduced by Caiani et al. (2019), with some adaptations required for readability and implementation purposes.

Subscripts originally used to identify economic entities are replaced as follows:

- z subscript is replaced by j subscript to identify banks;
- z subscript is used instead to identify any agent;
- CB subscript is replaced by n subscript to identify central banks;

To distinguish interest rate on different assets, superscripts is used systematically to identify these assets. For examples:

- r_{bt} is replaced by r_t^B to identify interest rate on bonds;
- r_{dt} is replaced by r_t^D to identify interest rate on deposits;
- r_{it} is replaced by r_t^L to identify interest rate on loans;
- r_{re} is replaced by r_t^R to identify interest rate on bank reserves;

Some stocks and flows variables have been replaced or introduced as follows:

- A is replaced by E to represent equities;
- A represent cash advances instead;
- Int represent interest paid;

- C for goods sales or consumption;
- WB for wages bill;

To simplify the notation of aggregate stocks or flows, we used:

- H subscript to identify collection of households;
- F subscript to identify collection of firms;
- B subscript to identify collection of banks;
- G subscript to identify collection of governments;
- CB subscript to identify collection of central banks.

These modifications only affect the representation of variables and do not modify the underlying assumptions, economic mechanism and accounting framework of the original model.

7.2. SFC accounting matrices

the SFC accounting matrices provide the accounting framework of the original model. This framework links the stocks identified in the section Entities, State Variables, and Scales, with the financial and real transactions and stocks represented in the original model. The following matrices present the representation associated with stocks entities and provide the accounting basis for tracking the stock and flow consistency of this replication. To build this accounting system which is implicit in the original model, we use work of Godley and Lavoie (2007) and Seppecher (2011)

7.2.1. Balance sheet matrix

The Balance sheet matrix represent assets and liabilities of each accounting sector. Accounting sectors correspond to group of agents identified in the model. Similarly, assets and liabilities correspond to monetary stocks identified in previous sections.

The Table 7 presents the balance sheet at any period. It is important to note that Loan entities are recorded in Advances position if they correspond to cash advances of Central Bank, while they are recorded in Loans position if they correspond to credit supplied by bank

Table 7. Aggregate Balance sheet of the economy

	Households	Firms	Banks	Government	C. Bank	Σ
Deposits	$+D_{Ht}$	$+D_{Ft}$	$-D_{Bt}$			0
Loans		$-L_{Ft}$	$+L_{Bt}$			0
Inventories		$+INV_{Ft}$				$+INV_{Ft}$
Bonds			$+B_{Bt}$	$-B_{Gt}$	$+B_{CBt}$	0
Reserves			$+R_{Bt}$	$+R_{Gt}$	$-R_{CBt}$	0
Advances			$-A_{Bt}$		$+A_{CBt}$	0
Equities	$+E_{Ht}$	$-E_{Ft}$	$-E_{Bt}$			0
Balance	$-NW_{Ht}$	$-NW_{Ft}$	$-NW_{Bt}$	$-NW_{Gt}$	0	$-INV_{Ft}$
Σ	0	0	0	0	0	0

The matrix of our balance sheet must follow essentially one single rule to be consistent: all the columns and all the rows that deal with financial assets or liabilities must sum to zero. The only row that may not sum to zero is the row dealing with inventories. indeed, the actual inventories accumulated by the firms are tangible real asset. A tangible asset or a real asset only appears in a single entry of the sectoral balance sheet, that of its owner. This is in contrast to financial assets and all liabilities, which are a claim of someone against someone else (Godley and Lavoie, 2007).

7.2.2. Transactions Matrix

The transaction matrices represent the monetary and real transactions occurring between agents during each simulation period. They identify the counterparties involved in each transaction and record the corresponding changes in their stocks and flows. By linking each transaction to the appropriate debit and credit entries across agents, these matrices provide the operational accounting framework through which individual transactions are recorded and ensure that the model preserves stock-flow consistency throughout the simulation

Table 8. Simplified aggregated transactions matrix of the economy

	Households	Firms	Banks	Government	C. bank	Σ
Consumption	$-C_{Ht}$	$+C_{Ft}$				0
Wages	$+WB_{Ht}$	$-WB_{Ft}$				0
Pub. transfers	$+G_{Ht}$			$-G_{Gt}$		0
Taxes	$-T_{Ht}$	$-T_{Ft}$	$-T_{Bt}$	$+T_{Gt}$		0
Dep. interests	$+Int_{Ht}^D$	$+Int_{Ft}^D$	$-Int_{Bt}^D$			0
Loan interests		$-Int_{Ft}^L$	$+Int_{Bt}^L$			0
Bond interests			$+Int_{Bt}^B$	$-Int_{Gt}^B$	$+Int_{CBt}^B$	0
Res. interests			$+Int_{Bt}^R$		$-Int_{CBt}^R$	0
Adv. interests			$-Int_{Bt}^A$		$+Int_{CBt}^A$	0
Dividends	$+Div_{Ht}$	$-Div_{Ft}$	$-Div_{Bt}$			0
C. Bank profit				$+\pi_{Gt}$	$-\pi_{CBt}$	0
Δ Deposits	$-\Delta D_{Ht}$	$-\Delta D_{Ft}$	$+\Delta D_{Bt}$			0
Δ Loans		$+\Delta L_{Ft}$	$-\Delta L_{Bt}$			0
Δ Bonds			$-\Delta B_{Bt}$	$+\Delta B_{Gt}$	$-\Delta B_{CBt}$	0
Δ Reserves			$-\Delta R_{Bt}$	$-\Delta R_{Gt}$	$+\Delta R_{CBt}$	0
Δ Advances			$+\Delta A_{Bt}$		$-\Delta A_{CBt}$	0
Δ Equities	$-\Delta E_{Ht}$	$+\Delta E_{Ft}$	$+\Delta E_{Bt}$			0
Σ	0	0	0	0	0	0

8. References

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